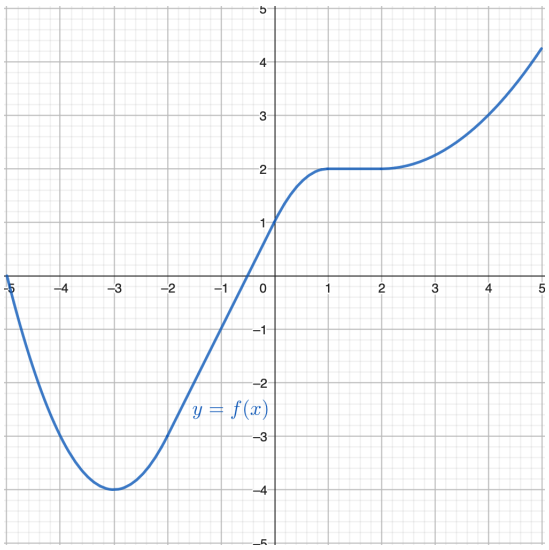
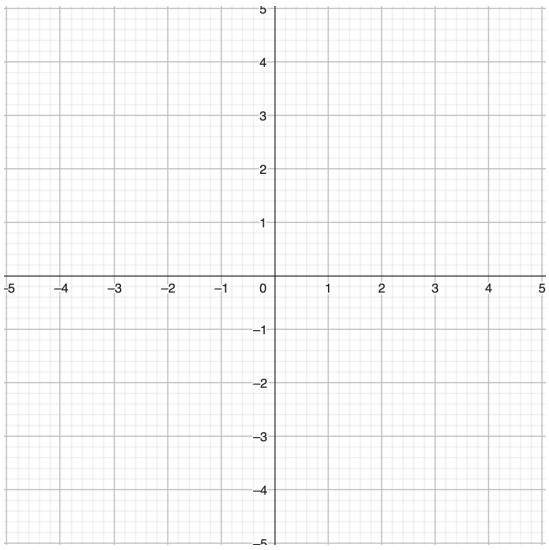
MATH 202 Your name: _____

Learning target C1, version 1

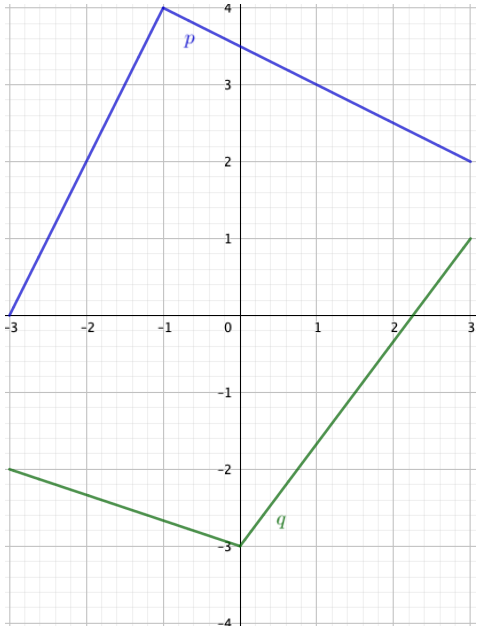
1. For the function f plotted at left, sketch an accurate graph of its derivative, f', on the axes at right.





2. Consider the two piecewise linear functions p and q given by the figure below.

   Determine each of the following derivative values that involve functions that are defined in terms of p and q. Clearly show your work and thinking using proper notation. (If I were you, I'd write out formulas for $m'(x)$ and $p'(x)$ before plugging in numbers.)



(i.) Let $m(x) = q(p(x))$. Determine $m'(-2)$.

(ii.) Let $n(x) = p(x^2)$. Determine $n'(-1)$.

Turn page over now please :)

3. Consider $\int \frac{1}{x}\,dx$.

   (a) What does this mean, in human words?

   (b) The antiderivative power rule won't work for this function. Why not? What goes wrong?

   (c) What's the actual answer to this question? (This is not a trick question. There *is* a good answer; it's not "undefined" or anything like that.)

   (d) Write some human words explaining why your answer is right.

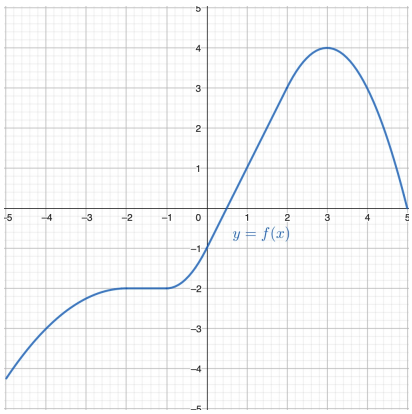
4. Evaluate the following definite integral exactly using the Fundamental Theorem of Calculus. Show your full work and thinking using proper notation.

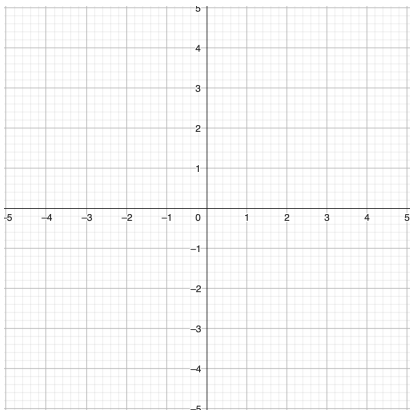
   $$\int_1^3 \left(4x^2 + \frac{1}{x^2} - \frac{2}{x}\right) dx$$

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Learning target C1, version 2

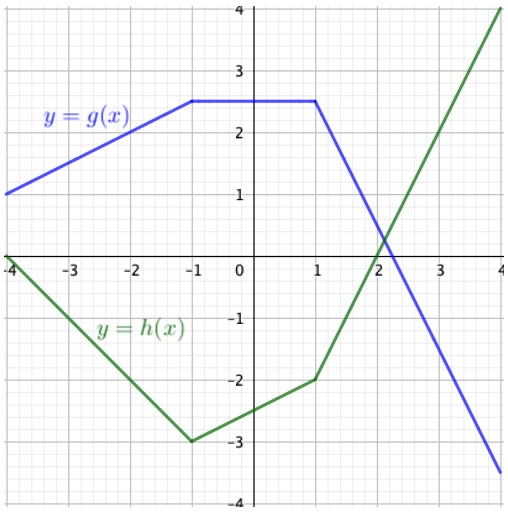
1. For the function f plotted at left, sketch an accurate graph of its derivative, f', on the axes at right.





2. Consider the two piecewise linear functions g and h given by the figure below.

   Determine each of the following derivative values that involve functions that are defined in terms of g and h. Clearly show your work and thinking using proper notation, be sure your values are *exact* in light of the given information, and write one sentence to explain your overall approach.



   (i.) If $N(x) = h(g(x))$, find $N'(x)$.

   Then, find $N'(3)$.

   (ii.) If $R(x) = h(x)^4$, find $R'(x)$.

   Then, find $R'(-2)$.

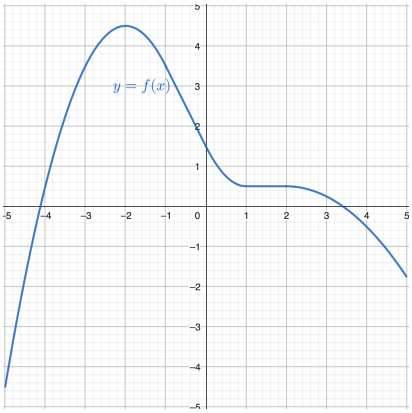
3. Evaluate the following definite integral exactly using the Fundamental Theorem of Calculus. Show your full work and thinking using proper notation.

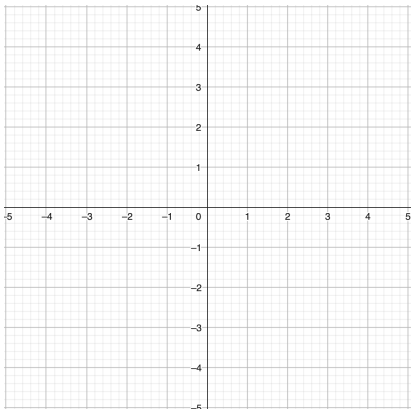
$$\int_1^2 \left(3x - e^x + x^{-1}\right) dx$$

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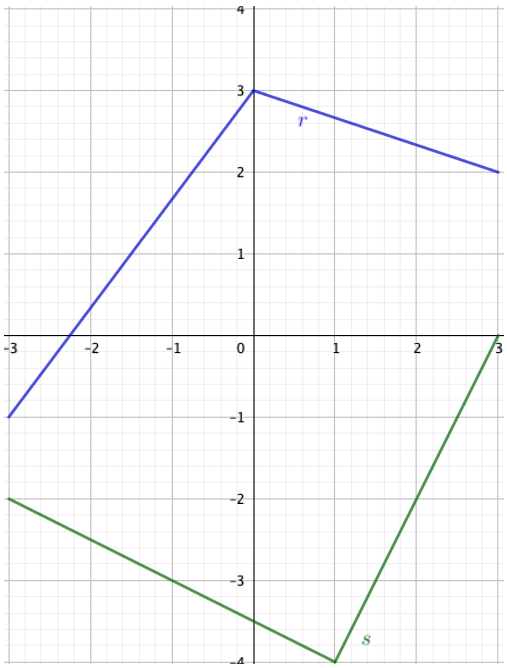
Learning target C1, version 3

1. For the function f plotted at left, sketch an accurate graph of its derivative, f', on the axes at right.





2. Consider the two piecewise linear functions r and s given by the figure below. Determine each of the following derivative values that involve functions that are defined in terms of r and s. Clearly show your work and thinking using proper notation, be sure your values are *exact* in light of the given information, and write one sentence to explain your overall approach.



(i.) If $M(x) = s(r(x))$, find $M'(2)$.

(ii.) If $T(x) = r(x^3)$, find $T'(-1)$.

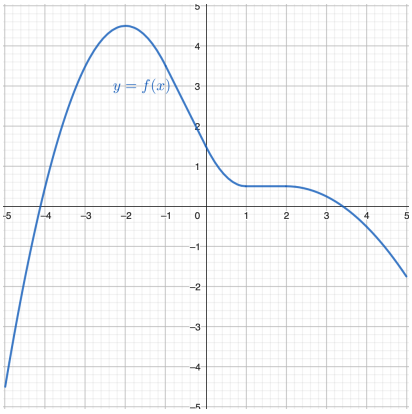
3. Evaluate the following definite integral exactly using the Fundamental Theorem of Calculus. Show your full work and thinking using proper notation.

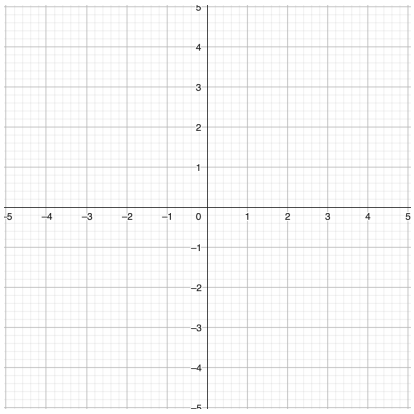
$$\int_{\pi/2}^{\pi} \left(\sin(x) + 2e^x - \frac{1}{x} \right) dx$$

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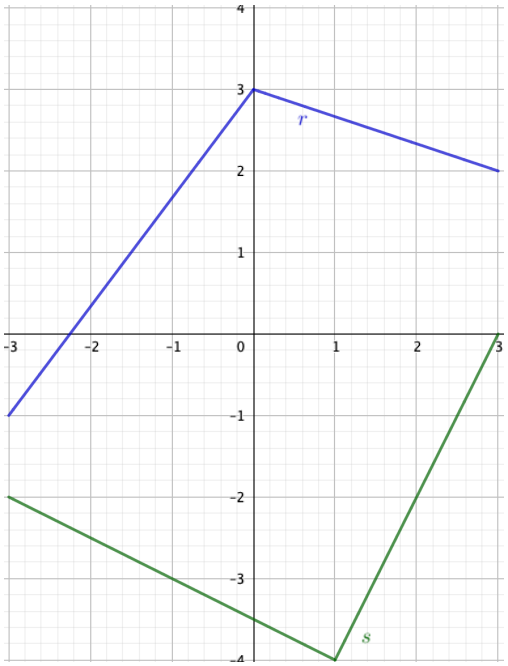
Learning target C1, version 4

1. For the function f plotted at left, sketch an accurate graph of its derivative, f', on the axes at right.





2. Consider the two piecewise linear functions r and s given by the figure below. Determine each of the following derivative values that involve functions that are defined in terms of r and s. Clearly show your work and thinking using proper notation, be sure your values are *exact* in light of the given information, and write one sentence to explain your overall approach.



(i.) If $Q(x) = s(r(x))$, find $Q'(x)$.
Then, find $Q'(2)$.

(ii.) If $W(x) = r(x^3)$, find $W'(x)$.
Then, find $W'(-1)$.

3. Evaluate the following definite integral exactly using the Fundamental Theorem of Calculus. Show your full work and thinking using proper notation.

$$\int_{\pi/2}^{\pi} \left(3 \cdot 2^x - \frac{2}{x} + \cos(x)\right) dx$$